

Scalability & Future Plan

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Model trained on limited historical weather data	May reduce prediction accuracy in unseen conditions	High
2	Manual weather data entry	Slower predictions and possibility of human error	Medium
3	Supports flood prediction only	Does not predict flood severity or affected areas	Medium

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users	Deploy on IBM Cloud with load balancing and auto-scaling
2	Data Storage	Stores historical weather data locally	Use cloud database for secure and scalable storage
3	Performance	Predictions generated for one request at a time	Optimize model and enable parallel processing with caching
4	Security	Basic Flask application security	Add authentication, HTTPS, encryption, and role-based access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate real-time weather APIs for automatic data collection	3–6 Months	Faster and more accurate predictions

Phase 3	Add GIS map visualization and SMS/ Email flood alerts	6–9 Months	Improved emergency response and public awareness
Phase 4	Deploy nationwide on IBM Cloud with mobile application support	9–12 Months	Large-scale adoption and better disaster management